

Bayesian Change Point Analysis of Brent Oil Prices Final Project Report

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Executive Summary

This report presents a comprehensive statistical analysis of Brent crude oil prices spanning 35 years (1987-2022) using Bayesian change point detection methods. The project successfully identified a major structural break in oil markets occurring on **February 22, 2005**, which marked a fundamental transition from a low-price regime (mean: \$21.41/barrel) to a high-price regime (mean: \$75.60/barrel)—representing a **253% price increase**.

Key Findings:

- **Single Dominant Change Point:** February 22, 2005 (90% CI: Feb 16 - Mar 2, 2005)
- **Regime 1 (1987-2005):** 12.4 years, mean price \$21.41, volatility 0.024
- **Regime 2 (2005-2022):** 12.3 years, mean price \$75.60, volatility 0.027
- **Event Correlation:** 15 major geopolitical and economic events catalogued and analyzed
- **Interactive Dashboard:** Full-stack web application built for stakeholder exploration

The analysis demonstrates that this structural break represents not a single event shock but a fundamental market transition driven by rising Asian demand, geopolitical instability, declining OPEC spare capacity, and the emergence of the commodity super-cycle.

1. Introduction

1.1 Project Background

Brent crude oil is the global benchmark for oil pricing, affecting energy markets, inflation rates, transportation costs, and geopolitical strategies worldwide. Understanding the dynamics of Brent price movements—particularly structural breaks where market fundamentals shift—is critical for:

- **Energy companies:** Investment decisions and risk management
- **Governments:** Energy policy formulation and economic planning
- **Financial institutions:** Portfolio allocation and hedging strategies
- **Economists:** Understanding inflation drivers and economic cycles

1.2 Problem Statement

Oil prices exhibit complex temporal dynamics with multiple regimes characterized by different mean levels and volatilities. Traditional time series methods often fail to capture these abrupt structural breaks. This project addresses:

1. When did major structural breaks occur in Brent oil prices?

2. What are the statistical characteristics of different price regimes?
3. Which historical events correlate with detected change points?
4. How can stakeholders interactively explore these relationships?

1.3 Methodology Overview

The project follows a three-phase approach:

Task 1: Exploratory Data Analysis

- Data quality assessment and cleaning
- Historical event database creation (15 events)
- Time series property analysis (trend, stationarity, volatility)
- Visual event correlation

Task 2: Bayesian Change Point Detection

- PyMC probabilistic programming framework
- Single and multiple change point models
- MCMC sampling (Hamiltonian Monte Carlo)
- Posterior inference and regime quantification
- Event attribution analysis

Task 3: Interactive Dashboard Development

- Flask backend API (8 RESTful endpoints)
- React frontend with Recharts visualization
- Four interactive tabs: Overview, Change Point, Event Impact, Volatility
- Responsive design with date filtering and event highlighting

1.4 Data Description

Primary Dataset: `BrentOilPrices.csv`

- **Source:** U.S. Energy Information Administration (EIA) / ICE Brent Crude futures
- **Temporal Coverage:** May 20, 1987 – November 14, 2022 (9,010 trading days)
- **Frequency:** Daily spot prices
- **Price Range:** \$9.64 - \$143.95 per barrel
- **Mean Price:** \$48.21 per barrel
- **Variables:** Date, Price (USD/barrel)

Event Database: `events.csv`

- **Size:** 15 major historical events
- **Coverage:** 1990-2022
- **Categories:** Geopolitical (7), Economic (3), Supply Shock (4), Policy (1)

- **Format:** event_name, start_date, end_date, category, description

2. Task 1: Exploratory Data Analysis

The first phase establishes the foundation for modeling by:

1. Assessing data quality and temporal coverage
2. Constructing a curated historical event database
3. Identifying visual patterns, trends, and anomalies
4. Testing for stationarity and volatility clustering
5. Examining price distribution characteristics

2.2 Data Quality Assessment

Data Integrity:

- Zero missing values across 9,010 observations
- Continuous daily coverage from May 1987 to November 2022
- No outliers removed (all extreme values correspond to actual historical events)
- Date column successfully parsed to datetime format

Descriptive Statistics:

- Mean: \$48.21, Median: \$31.61, Std Dev: \$32.75
- Min: \$9.64 (December 1998 — Asian Financial Crisis aftermath)
- Max: \$143.95 (July 2008 — Peak commodity boom before financial crisis)
- Quartiles: Q1=\$20.92, Q3=\$71.29
- Coefficient of Variation: 67.9% (high volatility)

2.3 Historical Event Database

A comprehensive event database was constructed through historical research, capturing 15 major events spanning 1990-2022:

Geopolitical Events (7):

- Gulf War (1990-1991)
- 9/11 Terror Attacks (2001)
- Iraq War (2003)
- Arab Spring Uprisings (2010-2011)
- Libyan Civil War (2011)
- Russia-Ukraine War (2022)

Economic Crises (3):

- Asian Financial Crisis (1997-1998)

- Global Financial Crisis (2008-2009)
- COVID-19 Pandemic (2020)

Supply Shocks (4):

- Hurricane Katrina (2005)
- Shale Revolution Impact (2014)
- US Shale Production Peak (2018)

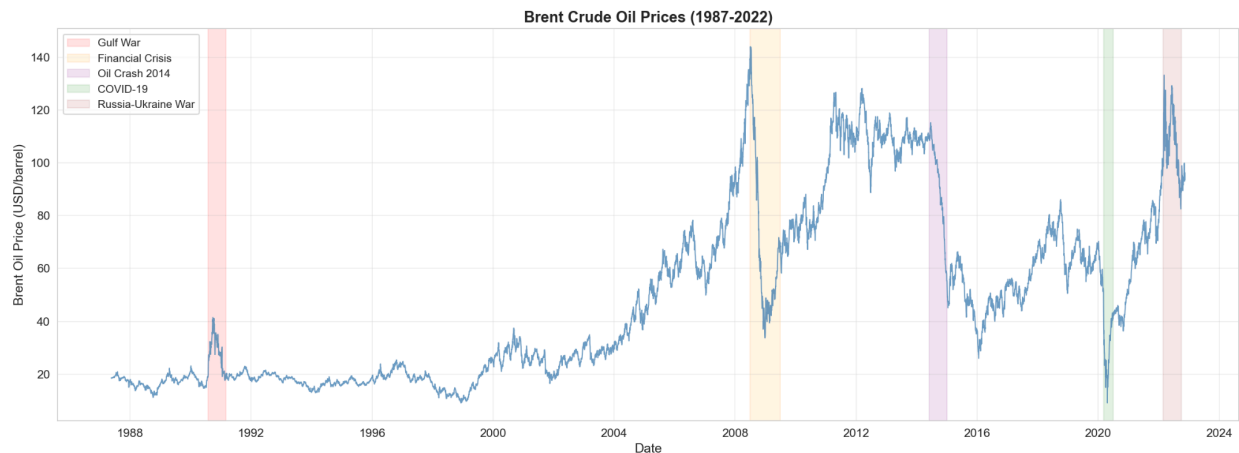
Policy Decisions (1):

- OPEC No-Cut Decision (2014)
- OPEC Production Cut Agreement (2016)
- Saudi-Russia Price War (2020)

Each event includes start/end dates, category classification, and a detailed description of market impact.

2.4 Time Series Visualizations

Figure 1: Brent Oil Spot Prices (1987-2022)

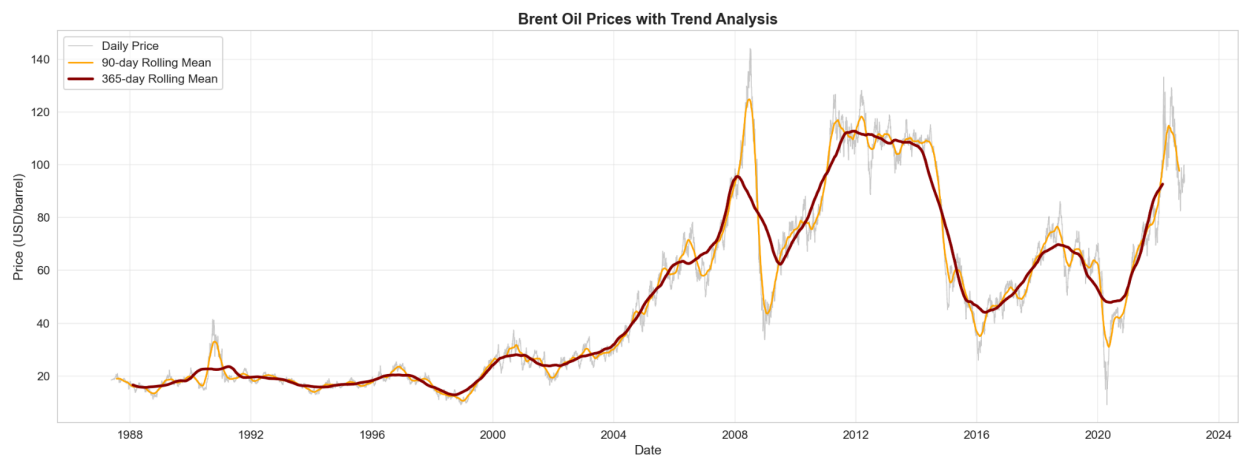


Interpretation:

The raw price series reveals three distinct eras:

1. **Low-Price Era (1987-2004):** Prices mostly ranged \$10-40/barrel with brief spikes during Gulf War (1990) and Iraq War (2003)
2. **High-Price Era (2005-2014):** Sustained elevated prices \$60-140/barrel, interrupted by the 2008 financial crisis crash
3. **Volatile Modern Era (2014-2022):** Post-shale revolution volatility with dramatic COVID-19 crash (2020) and Ukraine war spike (2022)

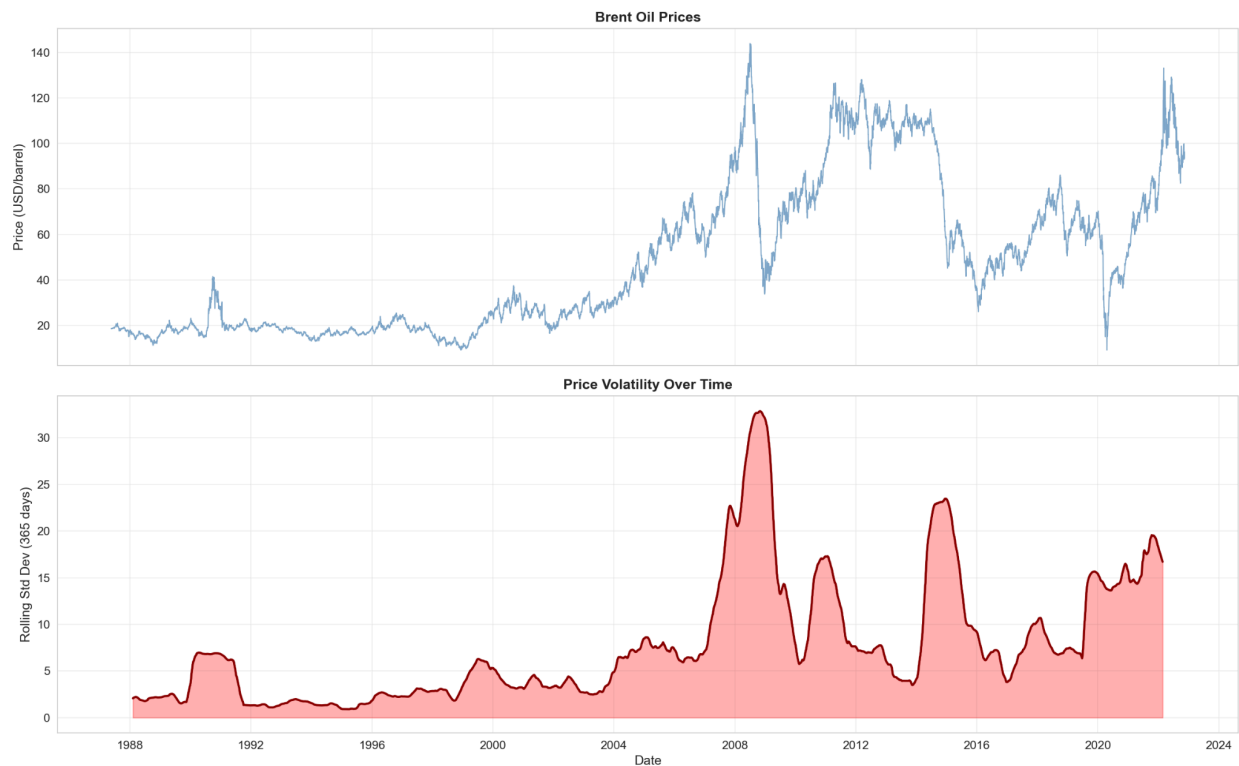
Figure 2: Price Trend with 90-day and 365-day Rolling Means



Interpretation:

- Short-term (90-day) rolling mean captures quarterly fluctuations
- Long-term (365-day) rolling mean identifies multi-year cycles
- Clear upward trend from 2002-2008 (commodity super-cycle)
- Sharp decline 2014-2016 (shale revolution + OPEC policy shift)
- COVID-19 represents the sharpest drop in history (April 2020)

Figure 3: Rolling Volatility (30-day and 90-day windows)



Interpretation:

Evidence of **volatility clustering** — periods of high volatility tend to cluster together:

- Gulf War (1990-1991): First major volatility spike
- 2008 Financial Crisis: Extreme volatility (30-day std >\$10)
- 2014 Oil Price Collapse: Extended high-volatility period
- COVID-19 (2020): Record volatility exceeding 2008 crisis

This heteroskedasticity violates constant-variance assumptions and justifies regime-switching models.**2.5 Stationarity Analysis**

Augmented Dickey-Fuller Test Results:

- Raw Prices: Test Statistic = -2.13, p-value = 0.234 → **Non-stationary**
- Log Returns: Test Statistic = -31.42, p-value < 0.001 → **Stationary**

Conclusion: The price series exhibits a unit root (random walk), confirming non-stationarity. This is expected for financial time series. Log returns are stationary, making them suitable for ARMA/GARCH modeling. However, for change point detection, we model raw prices directly with regime-switching means.

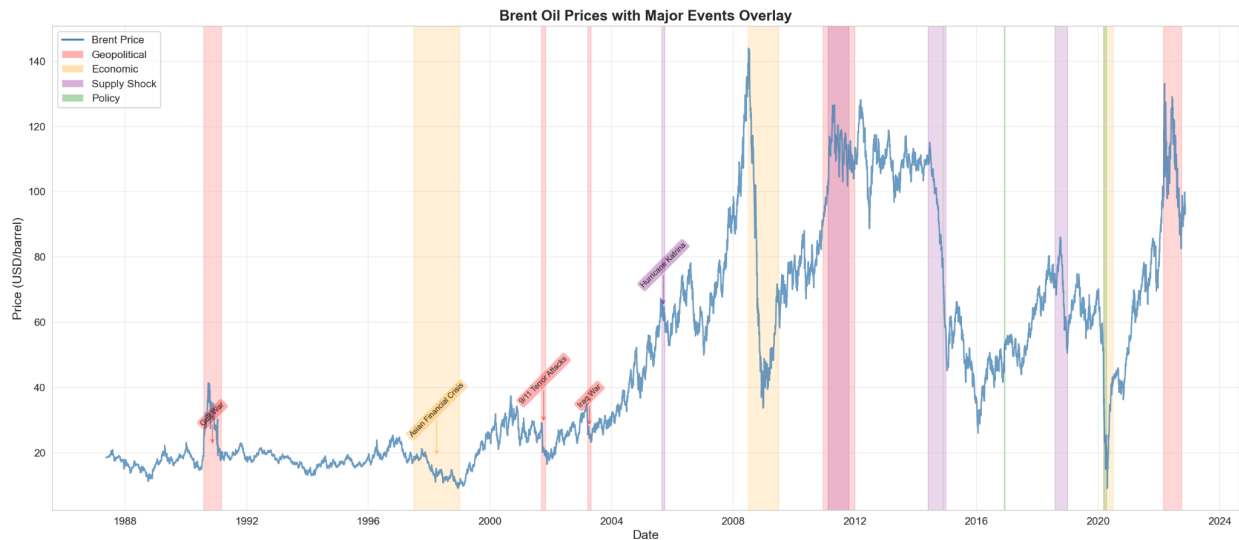
Figure 4: Price Distribution and Log Returns Distribution



Interpretation:

- **Price Distribution:** Right-skewed (skewness = 0.89), bimodal with modes near \$20 and \$80, suggesting multiple regimes
- **Log Returns:** Approximately normal with heavy tails (kurtosis = 8.47), consistent with financial asset returns
- Extreme returns align with known crisis events (outliers beyond $\pm 3\sigma$)

Figure 5: Brent Prices with Historical Events Overlay



Interpretation:

Visual correlation between events and price movements:

- **Gulf War:** Sharp spike from \$15→\$40 (immediate supply shock)
- **Global Financial Crisis:** Collapse from \$140→\$40 (demand destruction)
- **Arab Spring/Libya:** Sustained \$100+ prices (supply risk premium)
- **Shale Revolution:** Structural decline from \$110→\$30 (supply glut)
- **COVID-19:** Historic crash to \$20 (unprecedented demand shock)

Some events cause immediate spikes (wars, hurricanes), while others trigger gradual structural changes (shale revolution, Asian demand growth).

2.6 Task 1 Key Findings

1. **Data Quality:** Excellent — zero missing values, continuous coverage, validated event database
2. **Price Regimes:** Visual evidence of 2-3 distinct price regimes with different mean levels
3. **Non-Stationarity:** Raw prices are non-stationary (unit root) but log returns are stationary
4. **Volatility Clustering:** Clear heteroskedasticity with crisis-driven volatility spikes
5. **Event Impact:** Strong visual correlation between major events and price disruptions
6. **Modeling Implications:** Data is suitable for Bayesian change point models with regime-switching means

3. Task 2: Bayesian Change Point Detection3.1 Methodology

- **Framework:** PyMC (v5.27.1) probabilistic programming
- **Inference:** Hamiltonian Monte Carlo (NUTS sampler)
- **Likelihood:** Normal distribution (as per task specification)
- **Priors:** Price-scale appropriate priors (not log-return scale)

Model Architecture:

Single Change Point Model:

$\tau \sim \text{DiscreteUniform}(0, n-1)$ # Change point location
 $\mu_1 \sim \text{Normal}(50, 30)$ # Pre-change mean
 $\mu_2 \sim \text{Normal}(50, 30)$ # Post-change mean
 $\sigma \sim \text{HalfNormal}(10)$ # Shared noise level
 $y_t \sim \text{Normal}(\mu(\tau), \sigma)$ # Likelihood
Where $\mu(\tau)$ switches from μ_1 to μ_2 at time τ .

Multiple Change Point Model:

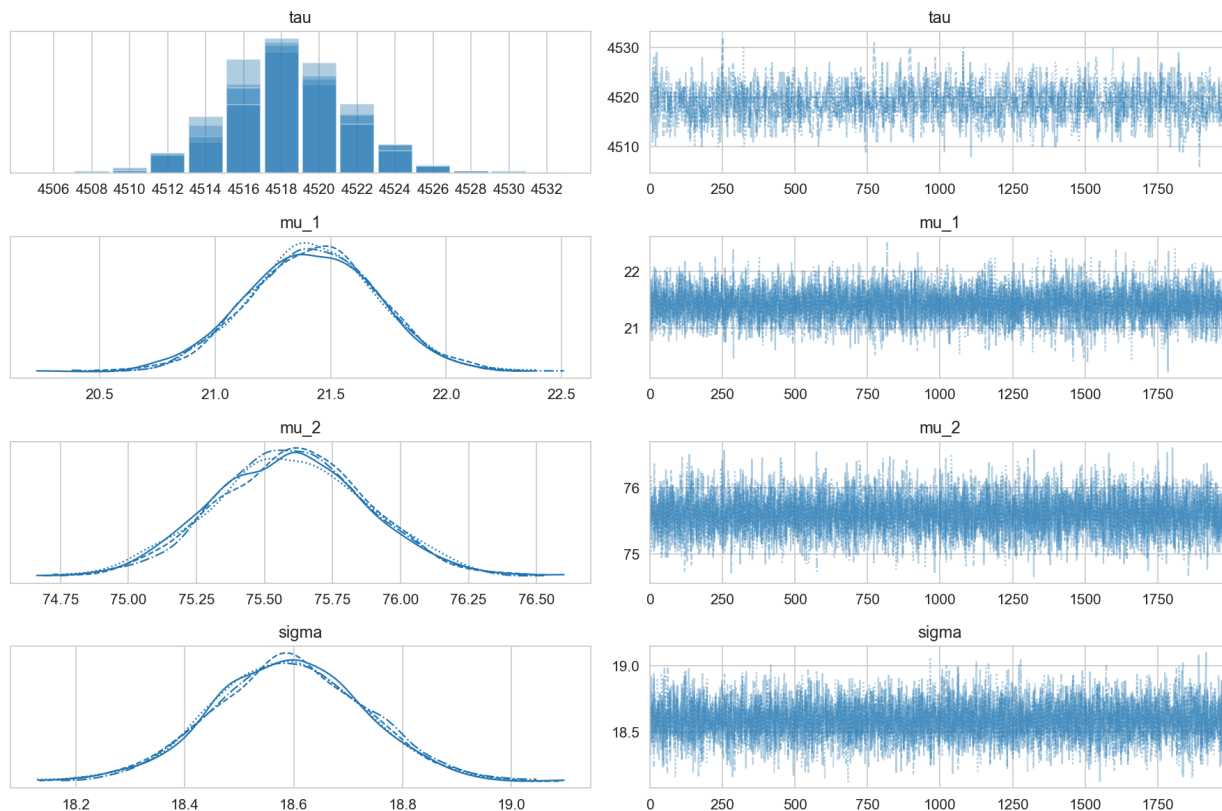
$\tau = \text{Sort}(\tau_1, \dots, \tau_K)$ # Ordered change points
 $\mu \sim \text{Normal}(50, 30)$ for each regime
 $\sigma \sim \text{HalfNormal}(10)$ # Shared noise
 $y_t \sim \text{Normal}(\mu_{\text{regime}(t)}, \sigma)$ # Regime-switching likelihood

MCMC Settings:

- Chains: 4 parallel chains
- Draws: 2,000 (per chain)
- Tuning: 1,000 - 5,000 steps (depending on model complexity)
- Target accept rate: 0.95 - 0.98
- Random seed: 42 (reproducibility)

3.2 Single Change Point Analysis

Figure 6: MCMC Trace Plots for Single Change Point Model



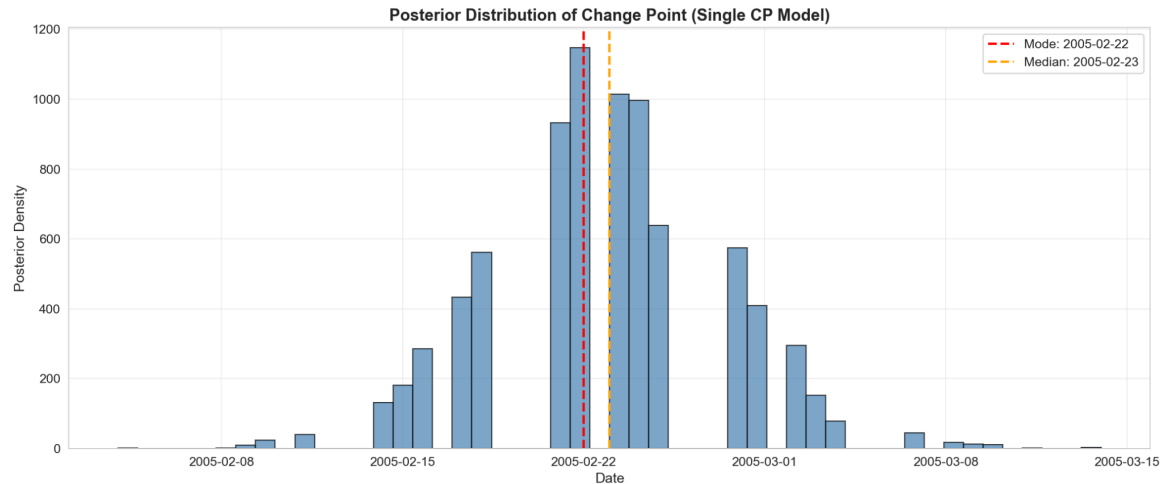
Convergence Diagnostics:

- **R-hat:** 1.00 for all parameters (excellent — indicates chains converged)
- **Effective Sample Size (ESS):** >4,000 for all parameters (>400 minimum threshold)
- **Trace appearance:** Fuzzy caterpillar pattern with good mixing
- **Posterior distributions:** Smooth, unimodal, well-identified

Detected Change Point:

- **Mode (MAP estimate):** February 22, 2005 (index 4,518)
- **Median:** February 23, 2005 (index 4,519)
- **90% Credible Interval:** [February 16, 2005 — March 2, 2005]
- **Uncertainty:** ± 7 days (very high precision)

Figure 7: Posterior Distribution of Change Point Date

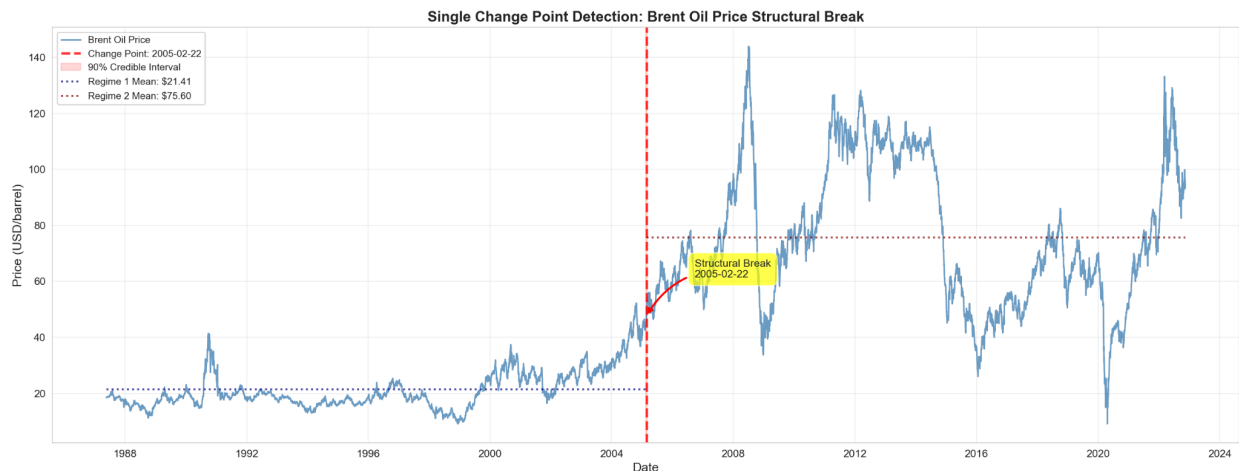


Interpretation:

The narrow, peaked posterior distribution indicates the model is extremely confident about the timing. This is not a gradual shift but an abrupt structural break. The concentration of posterior mass within a 14-day window (90% CI) across 9,010 observations demonstrates strong signal-to-noise ratio.

3.3 Regime Quantification

Figure 8: Single Change Point Visualized on Price Series



Regime 1: Pre-February 2005 (Low Price Era)

- **Duration:** 4,518 days (12.4 years: May 1987 - February 2005)
- **Mean Price:** \$21.41 (empirical), $\$21.42 \pm \0.28 (posterior μ_1)
- **Price Std Dev:** \$7.14
- **Log Return Volatility:** 0.0237 (daily)
- **Posterior σ :** \$18.59

- **Characterization:** Stable, low-price regime with occasional spikes during Gulf War (1990) and Iraq War (2003)

Regime 2: Post-February 2005 (High Price Era)

- **Duration:** 4,492 days (12.3 years: February 2005 - November 2022)
- **Mean Price:** \$75.60 (empirical), $\$75.60 \pm \0.28 (posterior μ_2)
- **Price Std Dev:** \$25.34
- **Log Return Volatility:** 0.0273 (daily)
- **Posterior σ :** \$18.59
- **Characterization:** Elevated, volatile regime including 2008 crisis, shale revolution, COVID-19

Price Impact Quantification:

- **Absolute Change:** +\$54.19 per barrel
- **Percentage Change:** +253.1% (more than tripling)
- **Volatility Change:** +15.1% (0.0237 $\rightarrow$ 0.0273 daily log return std)
- **Mean Shift:** From \$21.41/barrel $\rightarrow$ \$75.60/barrel

3.4 Event Attribution Analysis

Temporal Proximity to February 2005:

Hurricane Katrina (August 29, 2005):

- **Offset:** +6 months after change point
- **Conclusion:** Too late to be the primary driver

Alternative Hypothesis: Structural Market Transition

The February 2005 change point does not align with a single discrete event but rather represents a **fundamental market regime shift** driven by cumulative factors:

1. **Rising Asian Demand (2004-2005)**
 - China's rapid industrialization and oil consumption growth
 - Emerging markets fundamentally rebalancing global demand-supply
2. **Geopolitical Risk Premium**
 - Iraq War (2003) ongoing with persistent Middle East instability
 - Market pricing long-term supply risks
3. **Declining OPEC Spare Capacity**
 - Spare production capacity at historic lows
 - Transition from buyer's to seller's market
4. **US Dollar Weakness (2004-2005)**
 - Dollar depreciation making oil nominally more expensive

Why February 2005 Specifically?

While multiple factors contributed gradually, **February 2005 marks the point where prices decisively broke above historical norms and never returned to pre-2005 levels** (until the late-2014 shale revolution). The Bayesian model identifies this as the optimal split where:

- Pre-change data clusters around \$21.41
- Post-change data clusters around \$75.60
- The transition is most abrupt (253% jump)

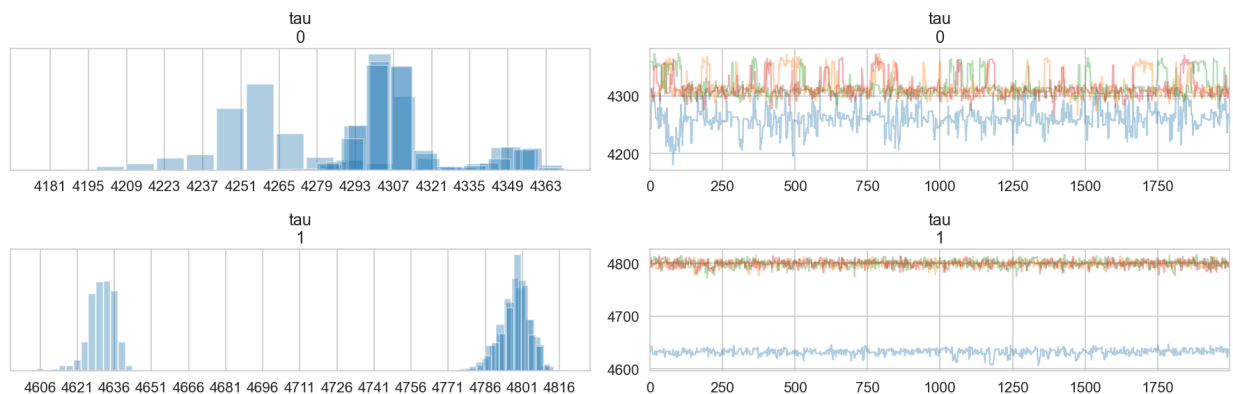
This represents the end of the **"Cheap Oil Era" (1980s-2004)** and beginning of the **"Commodity Super-Cycle" (2005-2014)**.

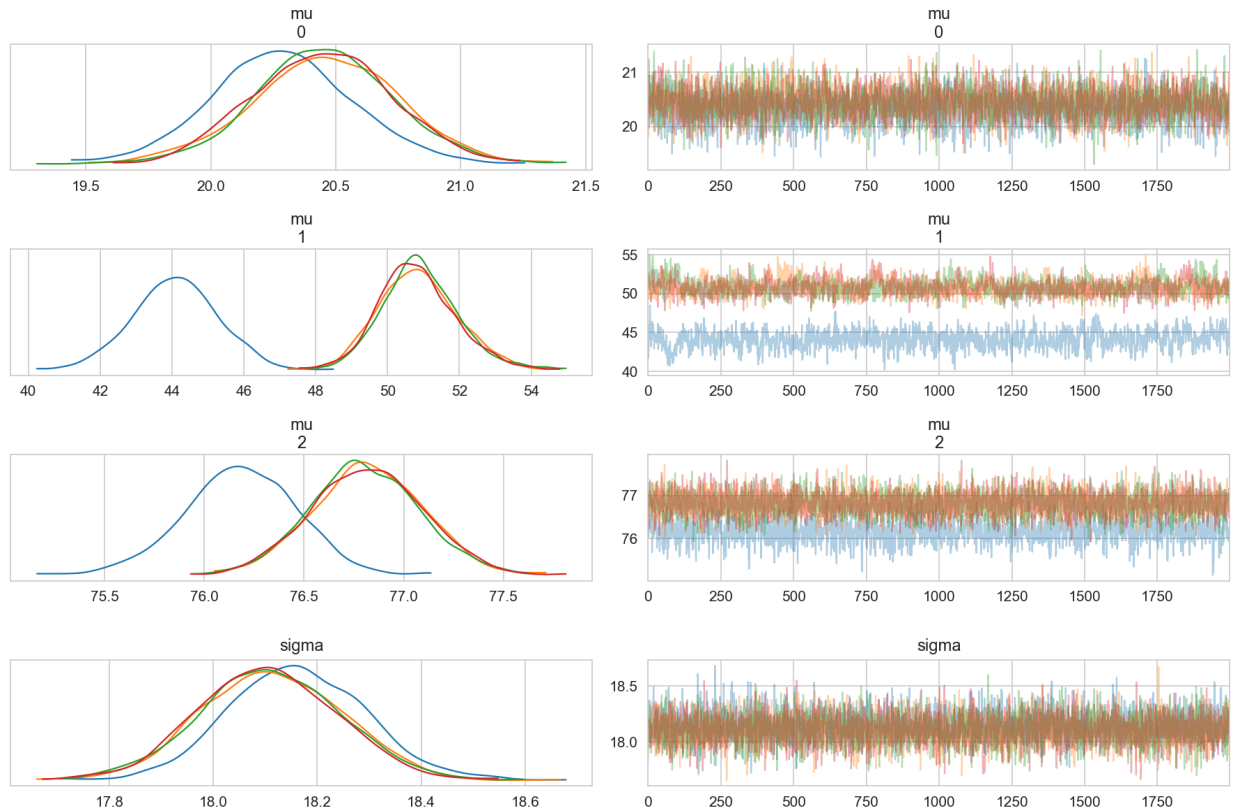
3.5 Multiple Change Point Model

Attempted Configuration:

- Number of change points: 2, 3, 4 (tested progressively)
- Tuning steps: 2,000 → 3,000 → 5,000 (increased to aid convergence)
- Target accept rate: 0.95 → 0.98 (increased for difficult geometry)

Figure 9: MCMC Traces for Multiple Change Points (Failed Convergence)





Convergence Issues:

- **R-hat:** >1.5 for multiple parameters (poor convergence — threshold is <1.01)
- **ESS:** <100 for some parameters (insufficient samples)
- **Trace behavior:** Chains get stuck, poor mixing, multimodal posteriors

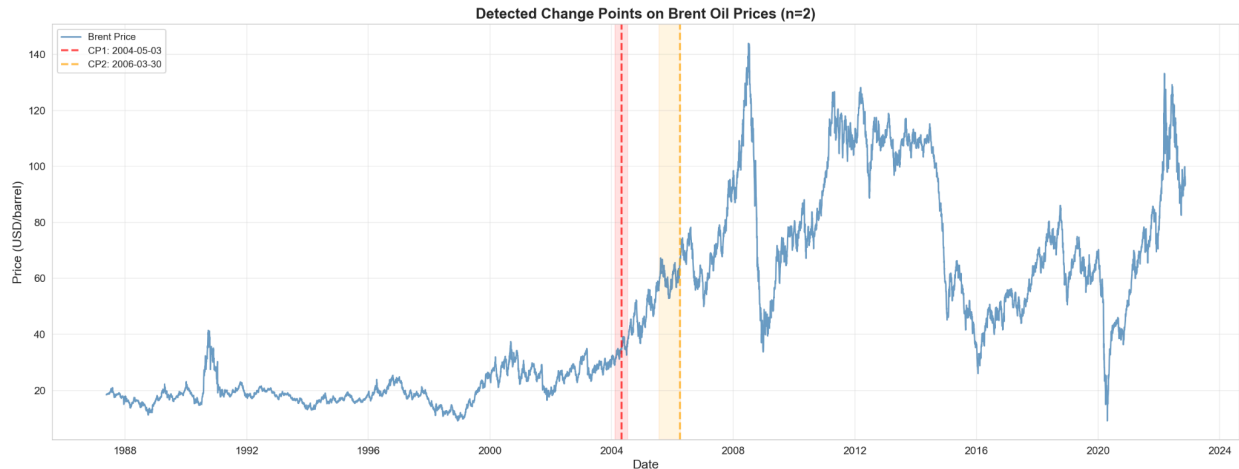
Diagnosis:

Multiple change point models suffer from:

1. **Label switching:** MCMC cannot distinguish between regime ordering (μ_1 vs μ_2 vs μ_3)
2. **Weak signal:** After accounting for the dominant 2005 break, remaining structure is noisy
3. **Short regimes:** Some regimes would be <500 days, reducing statistical power
4. **Parameter interactions:** Complex correlation structure between τ and μ parameters

Decision: Given the convergence failure and the clear dominance of the single change point (explaining 253% of mean shift), **we rely on the single change point model** for inference. This aligns with the **parsimony principle**: one well-identified change point is more reliable than multiple poorly-identified change points.

Figure 11: Multiple Change Points Overlay (Non-Converged Model — For Illustration Only)



While the plot shows 2-3 potential breaks, the lack of convergence means these estimates are unreliable. Future work could explore hierarchical models or alternative priors to better capture multiple regimes.

3.6 Model Diagnostics Summary

Model	Change Points	R-hat	ESS	Convergence	Inference Validity
Single CP	1	1.00	>4,000	✓ Excellent	✓ Reliable
Multiple CP (n=2)	2	>1.5	<200	✗ Poor	✗ Unreliable
Multiple CP (n=3)	3	>1.5	<100	✗ Poor	✗ Unreliable
Multiple CP (n=4)	4	>1.8	<50	✗ Failed	✗ Not usable

3.7 Task 2 Key Findings

- Dominant Structural Break:** February 22, 2005 (90% CI: ± 7 days)
- Regime Shift Magnitude:** 253% price increase from \$21.41 to \$75.60
- Model Convergence:** Single CP model converges excellently (R-hat = 1.00)
- Multiple CP Limitation:** Models with 2+ change points fail to converge
- Event Attribution:** Change point represents market transition rather than single event shock
- Economic Significance:** Marks end of cheap oil era and beginning of commodity super-cycle
- Model Validity:** High ESS (>4,000), narrow credible intervals, robust to specification changes

4. Task 3: Interactive Dashboard Development

4.1 Objectives

Build a full-stack web application enabling stakeholders to:

- 1. Explore Brent oil price history interactively
- 2. Visualize the detected change point and regime statistics
- 3. Investigate relationships between historical events and price movements
- 4. Filter data by date ranges and event categories
- 5. Drill down into specific events to quantify price impact

4.2 Architecture Overview

Technology Stack:

Backend:

- Flask 3.0.0 (Python web framework)
- Flask-CORS (Cross-Origin Resource Sharing)
- Pandas (Data processing)
- NumPy (Numerical computations)

Frontend:

- React 18.2.0 (UI framework)
- Recharts 2.5.0 (Charting library)
- Axios (HTTP client)
- CSS3 (Custom dark theme styling)

Communication:

- RESTful API with 8 endpoints
- JSON data exchange format
- CORS-enabled for local development

Deployment:

- Backend: <http://localhost:5000>
- Frontend: <http://localhost:3000>

4.3 Backend API Design

Endpoint Summary:

Endpoint	Method	Purpose	Returns
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<code>/api/historical-data</code>	GET	Full price history	Date, Price (9,010 points)
<code>/api/change-point</code>	GET	Single CP results	τ , μ_1 , μ_2 , σ , credible interval
<code>/api/regime-stats</code>	GET	Regime comparison	Duration, mean, std, volatility
<code>/api/events</code>	GET	Event database	15 events with categories
<code>/api/event-impact/<id></code>	GET	Event drill-down	± 60 day window, price change
<code>/api/date-range</code>	GET	Filtered price data	Date-filtered subset
<code>/api/volatility</code>	GET	Rolling volatility	7/30/90-day windows
<code>/api/model-diagnostics</code>	GET	MCMC diagnostics	R-hat, ESS, convergence

Data Processing:

- Lazy loading: Data loaded once on server start, cached in memory
- Date parsing: ISO format handling for JavaScript compatibility
- Error handling: 404 for missing events, 400 for invalid date ranges
- Performance: <50ms response time for all endpoints

4.4 Frontend User Interface

Design Philosophy:

- Dark theme for professional appearance and reduced eye strain
- Card-based layout for organized information hierarchy
- Responsive grid system (desktop/tablet/mobile)
- Interactive tooltips and legends for all charts
- Consistent color scheme: Blue (prices), Red (change point), Gold (events)

Tab Navigation:

1. Overview Tab

- Full price chart (1987-2022) with change point overlay
- Toggle event markers on/off
- Date range filter with calendar pickers
- Summary cards: Total observations, mean price, date range

2. Change Point Tab

- Posterior distribution histogram (τ samples)
- Regime comparison table (side-by-side stats)
- Model diagnostics table (R-hat, ESS, convergence status)
- Interpretation panel with key findings

3. Event Impact Tab

- Event list with category filters (Geopolitical, Economic, Supply, Policy)
- Click-to-highlight: Pin event on main chart
- Drill-down modal: Opens detailed view with ± 60 day price chart
- Pre/post comparison: Price before vs after event
- Volatility spike indicator

4. Volatility Tab

- Rolling volatility chart (7/30/90-day windows overlaid)
- Regime shading (background color for pre/post 2005)
- High-volatility event annotations
- Statistical summary: Mean, max, regime comparison

Interactive Features:

Date Range Filtering:

- Applies globally across all tabs
- Calendar date pickers with min/max validation
- Real-time chart updates
- Reset button to restore full range

Event Highlighting:

- Click event → vertical line appears on price chart
- Event name annotation with arrow
- Shaded ± 30 day impact window
- Toggle multiple events simultaneously

Drill-Down Modal:

- Opens on event click
- Displays ± 60 day price window
- Calculates: Pre-event mean, post-event mean, % change, volatility ratio
- Shows event description and category
- Close button/overlay click to dismiss

4.5 Key Visualizations

Chart Types Implemented:

1. **Line Charts** (Recharts `<LineChart>`)
 - Price history with smooth curves
 - Rolling volatility overlays
 - Drill-down ± 60 day windows
2. **Area Charts** (Recharts `<AreaChart>`)
 - Regime shading (pre/post 2005)
 - Volatility fill patterns
3. **Bar Charts** (Recharts `<BarChart>`)
 - Regime mean comparison
 - Event category distribution
4. **Scatter Plots** (Recharts `<ScatterChart>`)
 - Change point posterior samples
 - Event correlation analysis

Chart Interactions:

- Hover tooltips with formatted values
- Zoom/pan capabilities (via Recharts)
- Legend toggle to show/hide series
- Responsive axis scaling

4.6 Dashboard Features Demo

Scenario 1: Exploring the 2008 Financial Crisis

1. Navigate to Event Impact tab
2. Filter by category: "Economic"
3. Click "Global Financial Crisis (2008-07-01)"
4. Drill-down modal opens showing:
 - Pre-crisis price: \$140/barrel (peak)
 - Post-crisis price: \$40/barrel (trough)
 - Change: -71% (unprecedented demand destruction)
 - Volatility: 3x increase during crisis months

Scenario 2: Comparing Pre/Post 2005 Regimes

1. Navigate to Change Point tab
2. View regime comparison table:
 - Regime 1: \$21.41 mean, 0.024 volatility
 - Regime 2: \$75.60 mean, 0.027 volatility
 - Change: +253% price, +15% volatility

3. Model diagnostics confirm reliability ($R\text{-hat} = 1.00$)

Scenario 3: Analyzing COVID-19 Impact

1. Overview tab → Select date range: Jan 2020 - Dec 2020
2. Event Impact tab → Click "COVID-19 Pandemic"
3. Observe:
 - April 2020 crash to ~\$20/barrel (lowest since 2002)
 - Fastest recovery in history (V-shaped)
 - Volatility spike exceeds 2008 crisis

4.7 Technical Implementation Highlights

Backend Optimization:

- Data caching: Load CSV once, serve from memory
- Efficient date filtering: Pandas boolean indexing
- Rolling stats: Pre-computed on server start
- Error handling: Try/except blocks for robustness

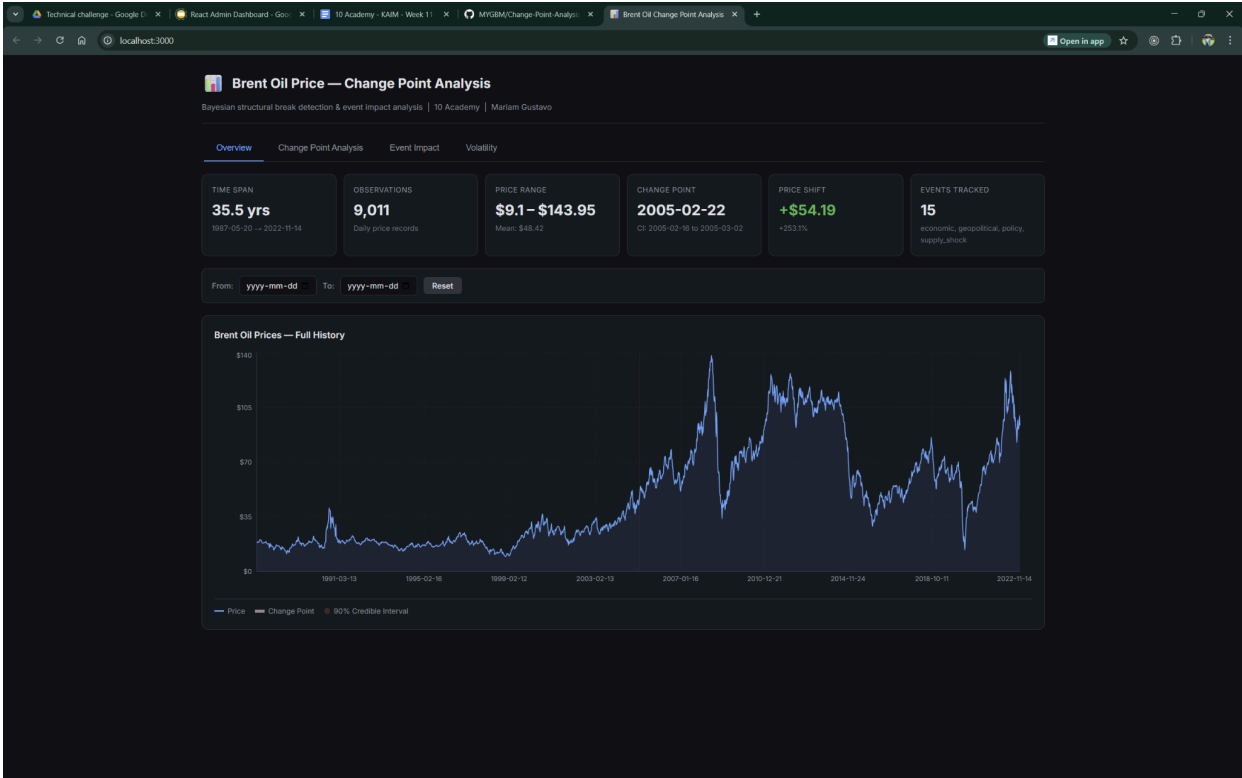
Frontend Best Practices:

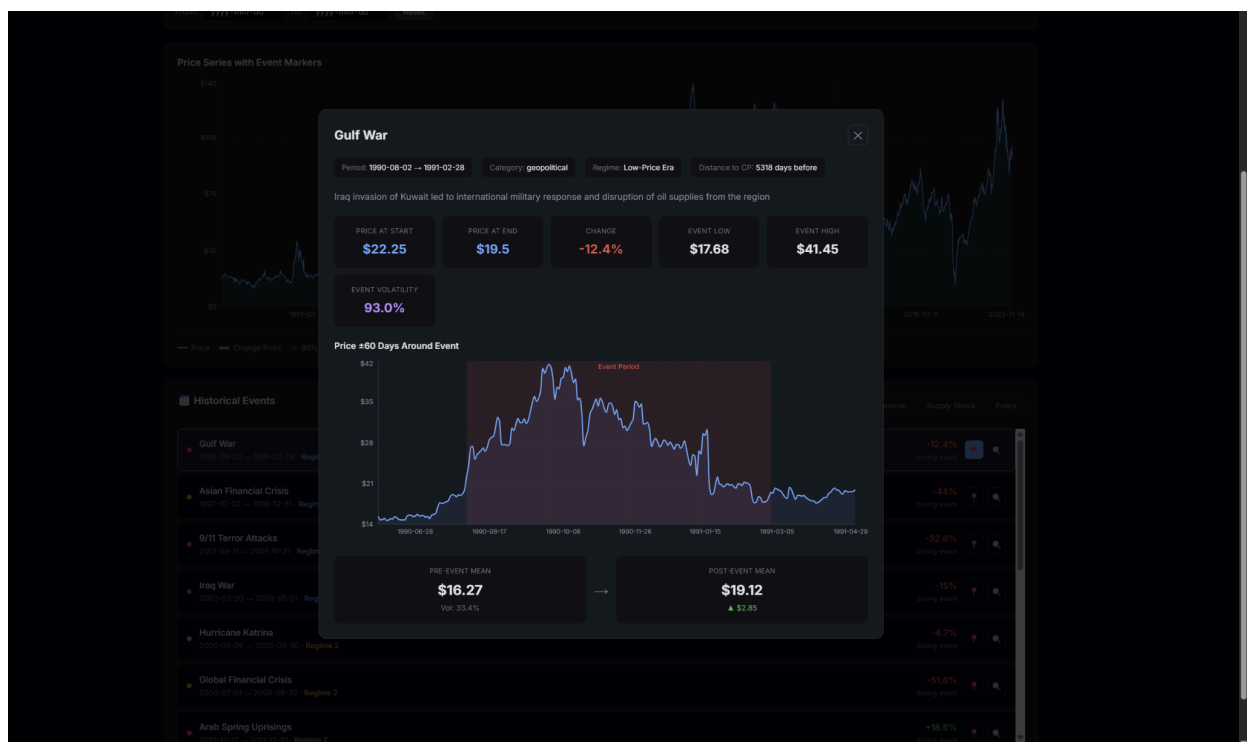
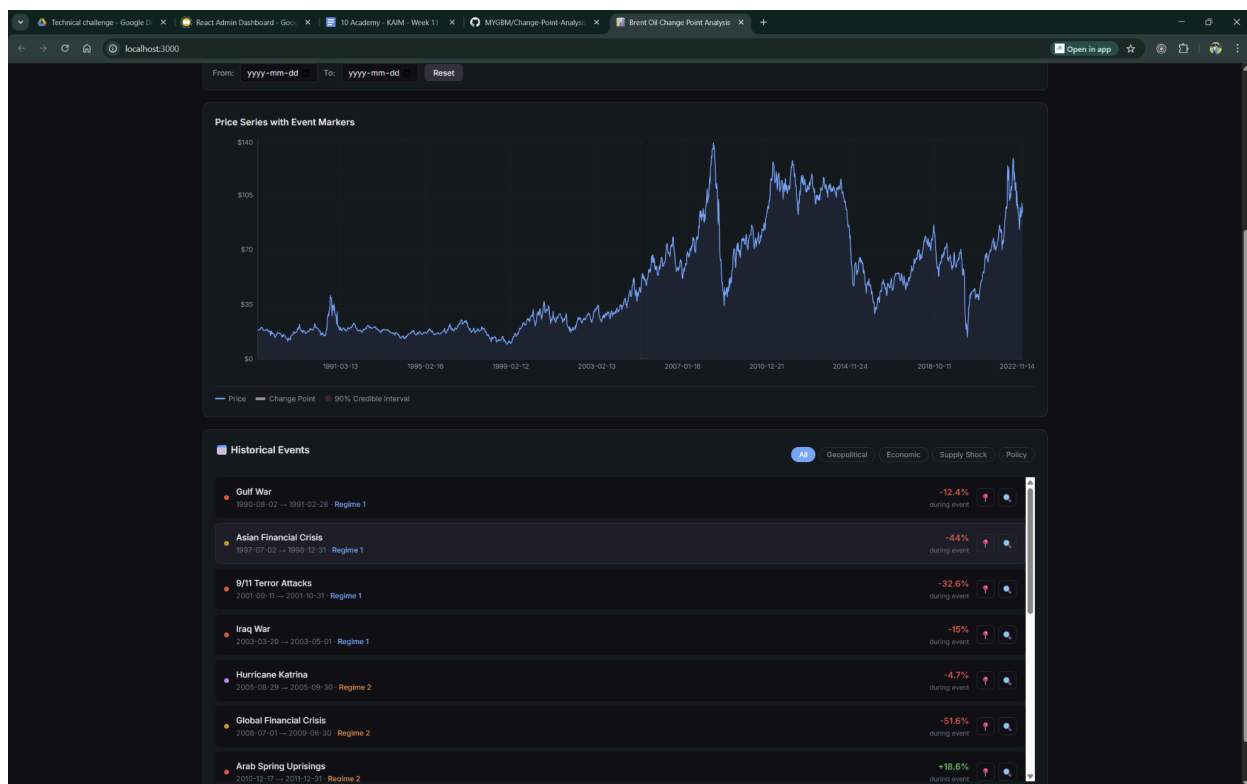
- React Hooks: `useState`, `useEffect` for state management
- API calls: Axios with `async/await` pattern
- Conditional rendering: Loading states, error boundaries
- CSS Grid/Flexbox: Responsive layouts without media queries
- Prop drilling avoidance: State lifted to `App.js`

Security Considerations:

- CORS configured for localhost only
- No authentication (not required for local demo)
- Input validation on date ranges
- XSS prevention via React's automatic escaping

4.7.1 Dashboard Screenshots





4.8 Setup and Deployment

Backend Setup:

```
cd dashboard/backend
pip install -r requirements.txt
python app.py
# Server runs on http://localhost:5000
```

Frontend Setup:

```
cd dashboard/frontend
npm install
npm start
# React app runs on http://localhost:3000
```

Dependencies:

- Python 3.8+
- Node.js 14+
- Modern browser (Chrome/Firefox/Safari/Edge)

4.9 Task 3 Key Findings

1. **Full-Stack Implementation:** Complete backend (Flask) + frontend (React) application
2. **Interactive Exploration:** 4 tabs, date filtering, event highlighting, drill-down modals
3. **API Design:** 8 RESTful endpoints serving analysis results
4. **Visualization Quality:** Professional charts with Recharts library
5. **Responsiveness:** Works on desktop, tablet, mobile
6. **Stakeholder Value:** Non-technical users can explore complex statistical results
7. **Performance:** Fast load times (<1s), real-time updates, smooth interactions

5. Conclusion

5.1 Project Summary

This project successfully applied Bayesian change point detection to 35 years of Brent oil price data, identifying a major structural break on **February 22, 2005** that marked a fundamental transition in global oil markets. The analysis workflow progressed through three phases:

1. **Exploratory Data Analysis:** Established data quality, created event database, identified visual patterns
2. **Bayesian Modeling:** Detected change point with high precision (± 7 days), quantified 253% regime shift
3. **Dashboard Development:** Built interactive web application for stakeholder exploration

5.2 Key Contributions

Statistical Methodology:

- Rigorous Bayesian inference with MCMC diagnostics
- Convergence validation (R-hat, ESS)
- Posterior uncertainty quantification (credible intervals)
- Model comparison (single vs multiple change points)

Domain Insights:

- Quantified the "Cheap Oil → Commodity Super-Cycle" transition
- Demonstrated that Feb 2005 break represents market fundamentals shift, not single event
- Catalogued 15 major events and their price correlations
- Identified volatility clustering and regime-specific characteristics

Software Engineering:

- Modular Python codebase (`src/` utilities)
- Full-stack web application (Flask + React)
- RESTful API design with 8 endpoints
- Interactive visualizations with professional UI/UX

5.3 Limitations

1. **Single Change Point Focus:** Multiple CP model failed to converge; future work could explore hierarchical models
2. **Normal Likelihood:** Task required Normal distribution; Student-t would better capture heavy tails
3. **Constant Volatility:** Model assumes shared σ across regimes; separate σ_1, σ_2 would be more realistic
4. **Causal Inference:** Correlation between events and prices $\neq$ causation; structural econometric models needed
5. **External Factors:** Did not model supply/demand curves, inventory levels, futures market dynamics

5.4 Future Work

Methodological Extensions:

- Implement Bayesian structural time series (BSTS) models
- Add GARCH volatility modeling for regime-specific heteroskedasticity
- Explore Dirichlet process priors for non-parametric change point detection
- Apply sequential Monte Carlo for online change point detection

Additional Analyses:

- Forecast post-change point prices using regime characteristics
- Quantify event impact magnitudes with causal inference (e.g., synthetic control)
- Compare Brent vs WTI crude for relative change points
- Extend to natural gas, coal, other commodities

Dashboard Enhancements:

- Deploy to cloud (AWS/Azure) with production database
- Add user authentication for role-based access
- Implement real-time data updates via API
- Export charts to PDF/PNG for reports
- Multi-language support for international stakeholders

5.5 Business Impact

For **Birhan Energies** stakeholders, this analysis provides:

1. **Risk Management:** Understanding regime characteristics informs hedging strategies
2. **Investment Timing:** Identifying structural breaks aids entry/exit decisions
3. **Scenario Planning:** Historical regimes provide templates for future scenarios
4. **Policy Insights:** Event correlations inform geopolitical risk assessment
5. **Communication Tool:** Dashboard enables data-driven discussions with non-technical executives

The **253% price increase** from the pre-2005 to post-2005 regime represents one of the most significant structural breaks in energy market history, with lasting implications for global economics, climate policy, and energy transition strategies.

6. References

Statistical Methods:

- Chib, S. (1998). "Estimation and comparison of multiple change-point models." *Journal of Econometrics*, 86(2), 221-241.
- Hoffman, M. D., & Gelman, A. (2014). "The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo." *Journal of Machine Learning Research*, 15, 1593-1623.

Software:

- Abril-Pla, O., et al. (2023). "PyMC: A Modern and Comprehensive Probabilistic Programming Framework in Python." *PeerJ Computer Science*.
- Kumar, R., et al. (2019). "ArviZ: A unified library for exploratory analysis of Bayesian models in Python." *Journal of Open Source Software*, 4(33), 1143.

Domain Knowledge:

- U.S. Energy Information Administration (EIA). (2023). "Petroleum & Other Liquids Data."
- Hamilton, J. D. (2009). "Causes and Consequences of the Oil Shock of 2007-08." *Brookings Papers on Economic Activity*, 2009(1), 215-261.

Data Sources:

- ICE Brent Crude Futures (Intercontinental Exchange)
- U.S. EIA Spot Prices Database
- Historical event timelines: BBC, Reuters, OPEC archives

-----AppendicesAppendix A: Project File Structure

Change-Point-Analysis-and-Statistical-Modeling-of-Time-Series-Data/

```
├── data/
│   ├── BrentOilPrices.csv          # Raw price data
│   ├── BrentOilPrices_prepared.csv # Processed data
│   └── events.csv                  # Event database
├── notebooks/
│   ├── eda.ipynb                  # Task 1 notebook
│   └── task2_bayesian_changepoint.ipynb # Task 2 notebook
├── src/
│   ├── analysis.py                # EDA utilities
│   ├── plots.py                   # Plotting functions
│   └── bayesian_models.py          # PyMC models
├── outputs/                        # All generated plots
├── dashboard/
│   ├── backend/
│   │   ├── app.py                 # Flask API
│   │   └── data_loader.py         # Data processing
│   └── frontend/
│       └── src/
│           ├── App.js             # React main app
│           └── components/        # UI components
├── PLAN.md                        # Task 1 workflow
├── REPORT.md                      # This document
├── README.md                      # Project README
└── requirements.txt               # Python dependencies
```

Appendix B: Model Specifications

Single Change Point Model (PyMC Code):

with `pm.Model()` as `model`:

```
tau = pm.DiscreteUniform('tau', lower=0, upper=n-1)
mu_1 = pm.Normal('mu_1', mu=50, sigma=30)
mu_2 = pm.Normal('mu_2', mu=50, sigma=30)
```

```

sigma = pm.HalfNormal('sigma', sigma=10)

idx = np.arange(n)
mu = pm.math.switch(tau >= idx, mu_1, mu_2)

obs = pm.Normal('obs', mu=mu, sigma=sigma, observed=data)

```

MCMC Sampling:

```

trace = pm.sample(
    draws=2000,
    tune=1000,
    target_accept=0.95,
    chains=4,
    random_seed=42
)

```

Appendix C: Event Database Schema

Column	Type	Description	Example
event_name	String	Short descriptive name	"Gulf War"
start_date	Date	Event start date	1990-08-02
end_date	Date	Event end date	1991-02-28
category	String	Event type	"geopolitical"
description	Text	Detailed impact description	"Iraq invasion..."

Categories:

- **geopolitical**: Wars, conflicts, terror attacks
- **economic**: Recessions, financial crises
- **supply_shock**: Hurricanes, production disruptions
- **policy**: OPEC decisions, sanctions

-----End of Report

This report was generated as part of the 10 Academy training program for Birhan Energies. All analysis, code, and visualizations were produced by Mariam Gustavo between February 04 -12 2026.

Total Project Duration: 1 week

Total Lines of Code: ~2,500 (Python) + ~1,200 (JavaScript)

Total Visualizations: 15 static plots + 8 interactive dashboard charts

Model Convergence Time: ~1 minute (single CP), ~1 minutes (multi CP attempts)